

Data Exploration and Cleaning

Data Analytics for Personalized Genomics and Precision Medicine

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The lecture follows a practical chain: obtain sequences, turn reads into a matrix, inspect data quality, and summarize the resulting measurements. These notes cover PDF pages 1–54. The main lesson is that reliable interpretation depends on both the data representation and the choices made during preprocessing.

1 Building a sequence from short reads

Lecture connection: PDF pp. 3–12.

1.1 Why alignment allows differences

A mismatch can reflect mutation, while insertions and deletions can create gaps. The recap illustrates a scoring matrix with positive match scores, negative mismatch scores and a gap penalty. Scoring rules encode how an alignment is evaluated; different matrices suit different biological contexts.

1.2 Overlap provides local evidence

The slides contrast reads of about 200 bases with a human genome of about 3 billion bases. Assembly connects overlapping reads to reconstruct longer sequences. For the toy reads TAA, AAT, ATG, TGC, GCC, CCA, CAT, ATG, TGG, GGA, GAT and ATG, one consistent result is TAATGCCATGGATG.

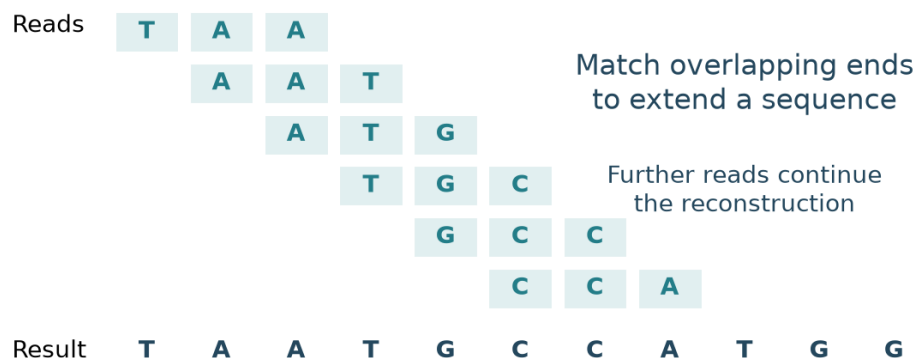


Figure 1. A selection of overlapping reads and the reconstructed sequence. Redrawn from the exercise on PDF p. 10.

Repeated motifs may fit at several locations; sequencing errors and conflicting bases make overlaps less certain. Repeated genes also make copy number difficult to determine. Assembly therefore involves deciding which connections are supported, as well as finding them efficiently.

2 Mapping reads and interpreting the matrix

Lecture connection: PDF pp. 14–26.

2.1 Locate reads on a known reference

Mapping begins with a reference and asks where each read fits. In the simplified exercise, a read is shifted one base at a time and mismatches are counted. A three-base read has 12 candidate starts in the 14-base reference. General alignment can accommodate gaps; the demonstration uses an ungapped comparison.

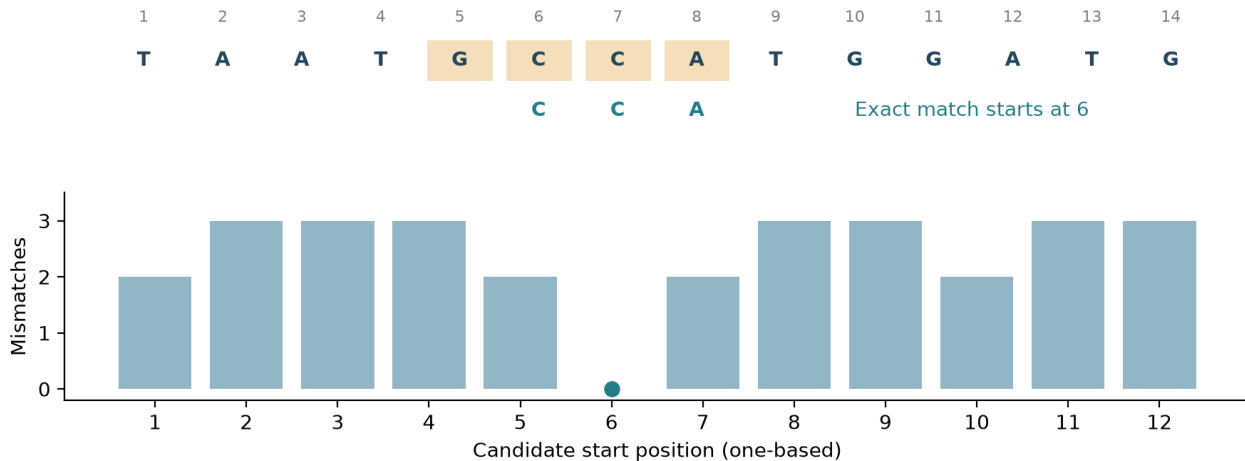


Figure 2. CCA matches positions 6–8 exactly. The shaded reference interval GCCA is the gene region illustrated in PDF p. 20.

For the highlighted gene, GCC and two CCA reads give the displayed count of 3. ATG has several exact placements, so a unique biological assignment cannot be inferred from that short sequence alone. Read counting requires a rule for ambiguous mappings and partial gene overlap.

2.2 Preserve the meaning of rows and columns

Operation	Available information	Result
Assembly	Reads and their overlaps	Reconstructed sequence
Mapping	Reads plus a reference	Candidate genomic locations
Counting	Assigned reads plus gene intervals	Gene counts for a sample

Repeating mapping and counting across samples yields a gene-by-sample matrix. In the general $n \times m$ matrix definition, rows are objects and columns are attributes. The four-person height-weight table is 4×2 ; person IDs are labels. Gene-expression tables may instead place genes in rows and samples in columns, so always read the axis labels.

Whole rows or columns can be reordered if labels move with them. Moving isolated cells breaks the association between measurements. Beyond matrices, the lecture lists sequential, spatial, temporal, graph, text and multimodal data; electronic health records may combine several of these forms.

Project idea on p. 23: obtain reference and read data, use mapping and counting software, and explain the inputs, outputs and purpose of each step.

3 Recognizing data quality problems

Lecture connection: PDF pp. 27–32 and 37.

Before choosing a treatment, identify the problem and the evidence for it. Exploration is useful here: unusual summaries or shapes can reveal issues that are hard to find in a large raw table.

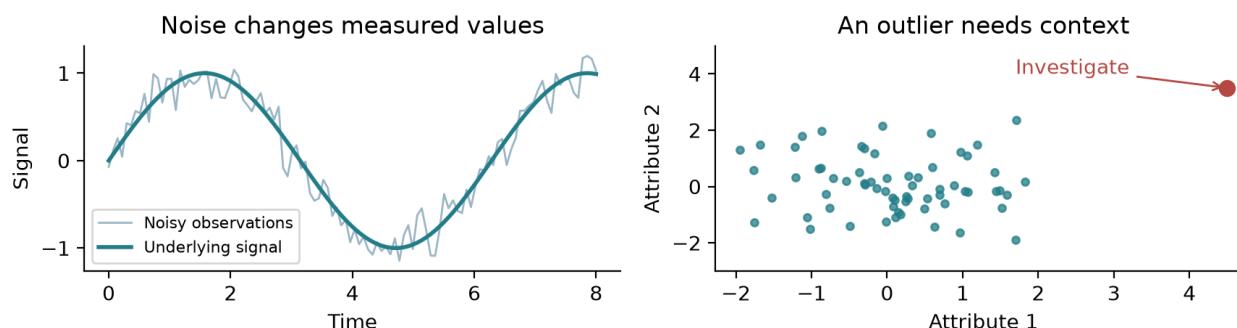


Figure 3. Original illustrative diagrams: noise around a signal and an isolated observation. The plotted values are synthetic, not lecture measurements.

Problem	Interpretation	Response to consider
Noise	Recorded values differ from the underlying signal.	Apply denoising only when a meaningful signal model or neighborhood exists.
Outlier	An observation differs markedly from the bulk of the data.	Investigate units, data entry and context before changing it.
Missing value	The value was not collected or the field does not apply.	Select a treatment that fits the reason for missingness.
Duplicate	The same observation is represented more than once.	Check identity and source records before retaining one copy.

3.1 Four approaches to missing values

Deleting an object removes the incomplete row but also loses its other measurements. Estimating a value retains the row but introduces an assumption. Ignoring missing entries can work when the calculation supports this. Considering possible values with probability weights represents uncertainty rather than pretending the missing value is known.

In the small height–weight table, height could guide a weight estimate, but similar height does not guarantee equal weight. Record which entries were imputed so later analysis can assess their influence.

3.2 Equal measurements do not establish duplication

Merging sources can introduce repeated records, as shown by P1 appearing in both example databases. Extra copies overrepresent that person in averages and frequency counts. Conversely, distinct people may legitimately share height and weight. Deduplication should use the identity of the observation, not numerical equality alone.

The slide checklist includes removing outliers. In practice, an unusual observation is a reason to investigate: it can be a valid and informative case. The chosen action should follow that investigation.

4 Scaling measurements and encoding categories

Lecture connection: PDF pp. 33–36.

4.1 Why measurement scales matter

For P1 = (1.79 m, 75 kg) and P2 = (1.64 m, 54 kg), the coordinate differences are 0.15 and 21. In squared Euclidean distance, their contributions are 0.0225 and 441. Thus the choice of units strongly affects an unscaled comparison.

Transformation	Formula and interpretation
Min-max scaling	$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$ The observed minimum becomes 0 and maximum becomes 1.
Z-score standardization	$z = (x - \text{mean}) / \text{standard deviation}$ The transformed value states distance from the mean in standard-deviation units.

A zero range or zero standard deviation makes the corresponding formula undefined. Such constant features require separate handling. Z-score scaling does not require Gaussian data to be computed and does not make an arbitrary distribution Gaussian; the slide's Gaussian assumption concerns how a standardized distribution may be interpreted.

4.2 Choose the direction deliberately

In the people table, scale height across people and weight across people. Combining height and weight within a row would mix unlike attributes. For a gene-by-sample matrix, scaling each gene across samples and scaling each sample across genes are different operations; the scientific question determines which is meaningful.

Person	Height (m)	Weight (kg)	Scaled height	Scaled weight
P1	1.79	75	0.625	0.875
P2	1.64	54	0	0
P3	1.70	63	0.250	0.375
P4	1.88	78	1	1

Worked check using PDF p. 34: P1 height = $(1.79 - 1.64) / (1.88 - 1.64) = 0.625$. These extrema belong to the four-person example; the merged exercise on the next page uses different values.

4.3 Give categories indicator columns

One-hot encoding represents a category by a 1 in its own column and 0 elsewhere. For columns (Red, Blue, Green), the codes are Red = (1,0,0), Blue = (0,1,0), and Green = (0,0,1). This avoids suggesting that one color is numerically larger than another. Categorical values are valid data; encoding is a representation choice for numerical methods.

5 Working through the cleaning exercise

Lecture connection: PDF p. 38. The following is an explicit worked solution, not a uniquely required answer.

Source	ID	Height (m)	Weight (kg)	Color
A	P1	1.79	75	Red
A	P2	1.64	Missing	Blue
A	P3	1.69	63	Green
A	P4	1.85	78	Red
B	P1	1.79	75	Red
B	P6	1.64	54	Green
B	P7	1.70	1000	Blue
B	P8	1.88	80	Green

Merge the tables and retain one P1 record. Flag 1000 kg as implausible and treat that weight as missing pending verification. As an illustrative imputation rule, use the observed weight at the nearest height: P2 receives P6's 54 kg, while P7 receives P3's 63 kg. Keep both imputations marked in the analysis record.

After these choices, height ranges from 1.64 to 1.88 m and weight from 54 to 80 kg. Scale the numerical columns separately and one-hot encode color. ID remains a label.

ID	Height scaled	Weight scaled	Red	Blue	Green
P1	.6250	.8077	1	0	0
P2	.0000	.0000	0	1	0
P3	.2083	.3462	0	0	1
P4	.8750	.9231	1	0	0
P6	.0000	.0000	0	0	1
P7	.2500	.3462	0	1	0
P8	1.0000	1.0000	0	0	1

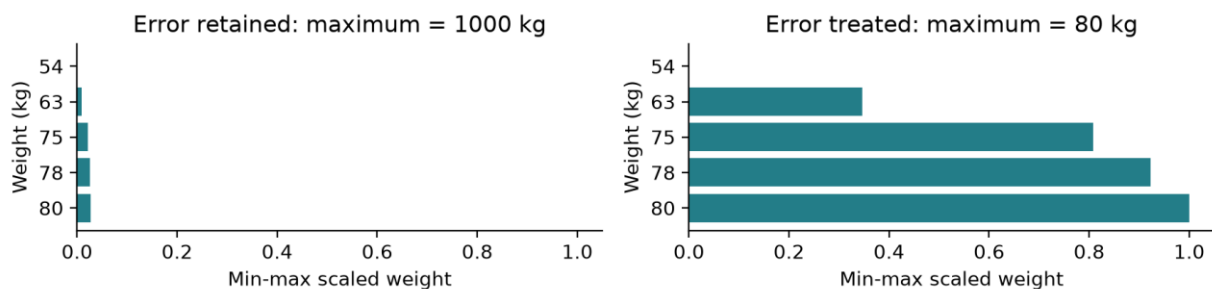


Figure 4. An erroneous maximum compresses valid weights. Both panels show the same five valid weights; the left scaling also uses the 1000 kg error.

For 75 kg, keeping the error gives $21/946 \approx 0.0222$; after treatment, the value is $21/26 \approx 0.8077$. Resolve the suspect extreme before fitting the scale. The table contains seven people and five numerical features.

6 Summarizing location and spread

Lecture connection: PDF pp. 41–44.

Summary statistics describe frequency, location and spread. A mean alone does not describe the whole distribution: two datasets may have the same center and very different variation.

6.1 Describe a typical value

The mean is the sum divided by the number of observations. The median is the middle ordered value, or the average of the two central values for an even sample. A trimmed mean removes a chosen fraction from each tail before averaging. The median is less affected by extremes than the mean.

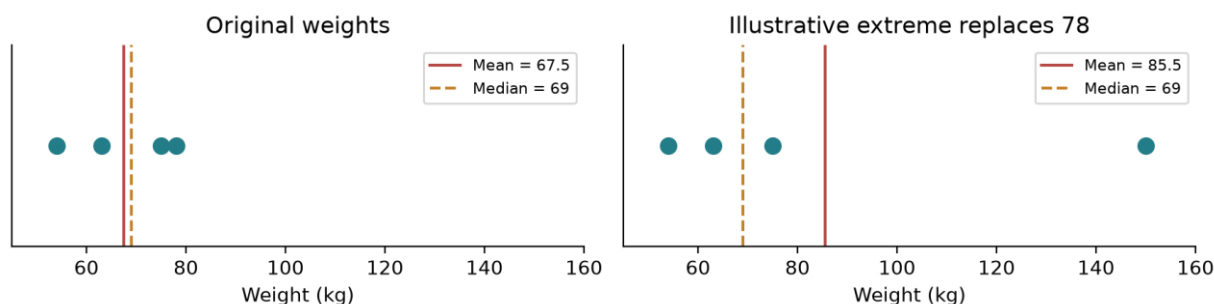


Figure 5. An illustrative replacement of 78 by 150 shifts the mean from 67.5 to 85.5 kg; the median stays at 69 kg.

6.2 Describe variation around the center

Measure	Definition
Range	Maximum – minimum
Sample variance	$s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$
Sample standard deviation	$s = \sqrt{s^2}$; expressed in the original measurement unit
Interquartile range	IQR = Q3 – Q1; width of the middle 50%
Median absolute deviation	MAD = median(x_i – median(x))

Using weights 54, 63, 75 and 78 kg: mean = $270/4 = 67.5$ kg; median = $(63 + 75)/2 = 69$ kg; range = 24 kg. The squared deviations total 369 kg², giving sample variance 123 kg² and sample standard deviation about 11.09 kg. Treating these four values as the entire population would instead give variance 92.25 kg².

For MAD, the absolute distances from 69 are 15, 6, 6 and 9 kg. Their median is 7.5 kg. This is the unscaled MAD. Definition note: PDF p. 44 prints deviations from the mean under the MAD label; the standard median absolute deviation uses the median as its center.

7 Percentiles and visual exploration

Lecture connection: PDF pp. 45–51.

7.1 Percentiles require a convention

A percentile locates a value within ordered observations. The slide describes the p th percentile using the fraction strictly below it. For finite datasets, ties and interpolation rules matter: in 1, 2, ..., 10, exactly 30% are below 4, while exactly 30% are at or below 3. These counting statements are different from a software interpolation rule, which may return a non-integer value.

State the convention when calculating quartiles or comparing software results. The 50th percentile describes the center; Q1 and Q3 correspond to the 25th and 75th percentiles.

7.2 Frequency and mode summarize categories

A count records how often a value occurs; relative frequency divides that count by the total. The mode is the most frequent value. In the cleaned seven-person example, Green has relative frequency $3/7 \approx 42.9\%$ and is the mode; Red and Blue each occur $2/7$ of the time.

7.3 A visual display reveals structure

Visualization converts measurements into a form where patterns, trends and unusual observations are easier to recognize. The sea-surface-temperature map on p. 49 combines geographic position with a color scale: longitude and latitude locate observations, while color conveys temperature. A map preserves spatial relationships that a long numerical list obscures.

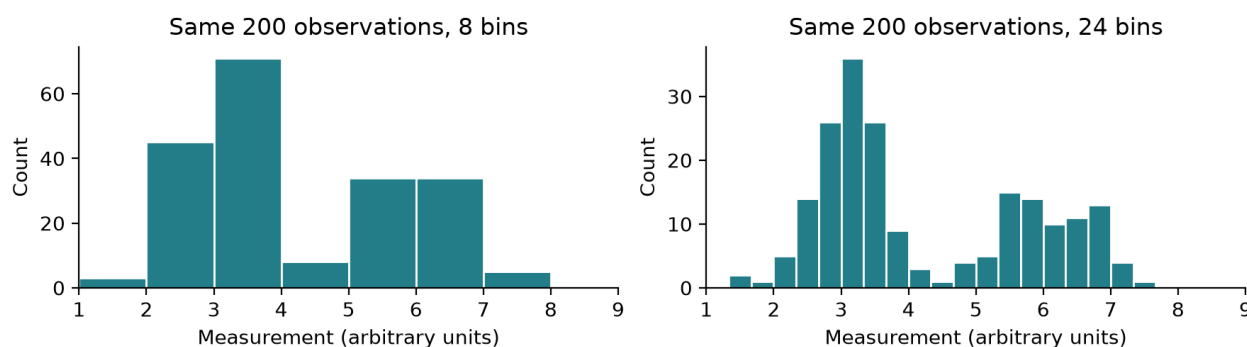


Figure 6. Original histogram example using synthetic measurements. Bin choice changes the visible detail even though the observations are unchanged.

A histogram divides one numerical variable into bins and counts observations in each bin. Few bins can hide structure; many bins can make random fluctuations prominent. Read both the horizontal measurement scale and the vertical count scale. A separated bar may flag an extreme value, but the plot alone cannot establish that it is erroneous.

Use plots and statistics together: a numerical summary quantifies a pattern, while the plot shows its shape and possible exceptions. Revisit both after cleaning.

8 Joint distributions and box plots

Lecture connection: PDF pp. 52–54.

8.1 Inspect two attributes together

A two-dimensional histogram bins both attributes and counts observations in each pair of intervals. PDF p. 52 uses petal length and width. A concentration along a diagonal can suggest association; separated concentrations can suggest groups worth investigating. Such patterns do not, by themselves, prove a causal relationship or establish group identities.

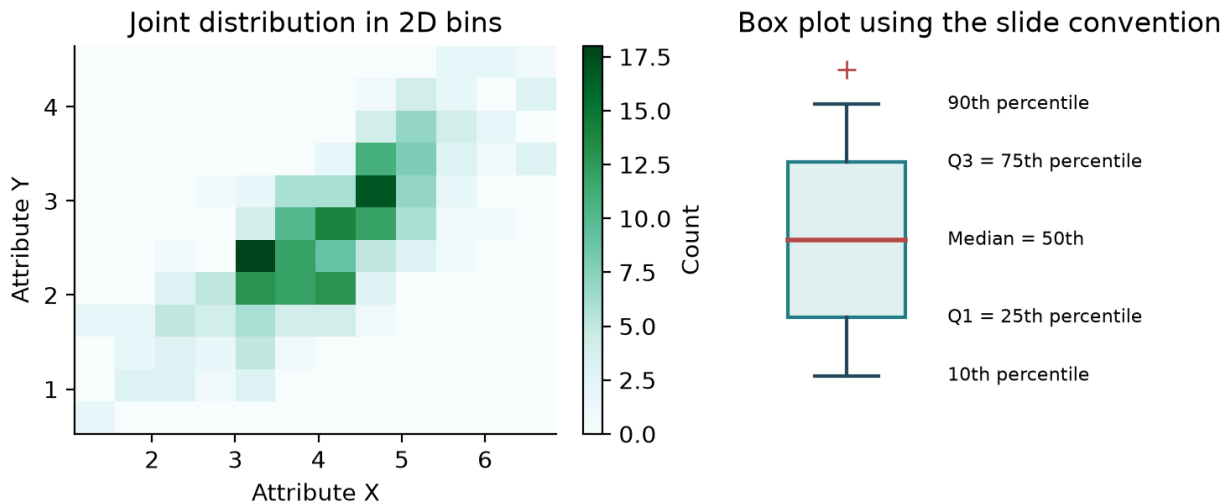


Figure 7. Left: a synthetic joint distribution encoded by cell color. Right: a schematic box plot reproducing the percentile-whisker convention labeled on PDF p. 53.

8.2 Read the box plot convention

The box spans Q1 to Q3, its height is the IQR, and the internal line marks the median. Comparing boxes reveals differences in center and spread. The slide explicitly labels whiskers at the 10th and 90th percentiles. Points beyond those limits are tail observations; they are not automatically errors.

Many programs instead place whiskers at the most extreme observations within $1.5 \times \text{IQR}$ of the box. Check the plotting convention before interpreting points outside the whiskers. The same data can have different whiskers under these two rules.

8.3 Questions to carry into the tutorial

What does each row represent? Which values were changed and why? Were scaling parameters calculated after investigating errors? Which percentile and whisker conventions are used? Do the plots support the interpretation suggested by the summary statistics? Page 54 introduces a Colab exploration example and further tutorial practice.

Source and figure notes

Format reference: supplied BMEG3105_Lec5_Scribing.docx. Its numbered sections, worked examples and explanatory captions informed the presentation. These notes reorganize the material and independently redraw the figures. Synthetic illustrations are identified in their captions; calculations use the slide values where specified.

Preparation: illustrated with OpenAI Codex assistance.